FX PB Collateral

Automating FX Prime Brokerage Monitoring

Replacing tedious manual portal extractions with an automated end-to-end data pipeline & Power BI analytics dashboard

The Business Problem & Enterprise Impact

Understanding the manual bottlenecks in daily FX Prime Brokerage collateral and limits reporting.

Manual Process Challenge

1

Portal Authentication

Operations team had to manually log into the Prime Brokerage (PB) portal every morning, navigate through reporting sub-menus, and await slow report loads.

2

Manual Screenshotting

Gather relevant quantitative and qualitative data from internal and external sources

3

Teams Pasting & Delays

Clean incomplete records, remove duplicates, and ensure accuracy to maintain data integrity

Implemented Solution

Automated Ingestion Pipeline

Automated Email Delivery

PB portal configured to auto-distribute daily CSV files

Power Automate Workflow

Cloud workflow automatically saves CSV files to a secured SharePoint folder

Power BI Dataflows

Connects to SharePoint, cleans, aggregates, and structures the raw CSV data for downstream reporting

Power BI Reports

Automatically refreshed interactive reports distributed to the primary stakeholders

Operational Speed & Impact

<10s

Time to Decision Insight

Instant Collateral Requirements Visibility

By removing manual portal logins, screenshotting, and Teams messaging, the solution compresses the daily reporting cycle from over 15 minutes down to under 10 seconds.

Stakeholders receive updated and reliable collateral report every morning.

Public Portfolio & GitHub Replication

Enterprise project securely recreated into an open-source demonstration

Enterprise vs Open Source

Workplace Production		GitHub Replication
Prime Brokerage portal CSV feeds	**Data**	Custom Python script generating synthetic FX data
Microsoft SharePoint Online	**Storage**	PostgreSQL on Neon Database
Power BI Dataflows to BI reports	**Connection**	Direct Power BI connection to Neon DB.
Power BI Report	**Reporting**	BI Tools – Power BI or Google Data Studio

Data Architecture

Python & Neon PostgreSQL Pipeline

Data Generation

Developed a Python script to synthesize mock data mimicking original reports

Database Integration

Structured a relational schema and automated data loads directly into a Neon PostgreSQL serverless cloud database

Power BI Reports

Connected Power BI Reports directly to Neon Database to model data, write DAX measures, and publish the dashboard

AI-Assisted Development

1

Rapid Scripting & Synthetic Logic

Leveraged Generative AI assistance to program complex Python generator scripts

2

Database Schema Optimization

Leveraged AI prompts to help design PostgreSQL tables— schema layout, primary keys, and data types for Neon DB

3

Documentation

Generated docstrings and a comprehensive GitHub README explaining setup instructions for reviewers

4

Implementation

AI accelerated implementation while designing, testing, and validating the logic.

Reporting Examples

Interactive Power BI Reports

Collateral Dashboard